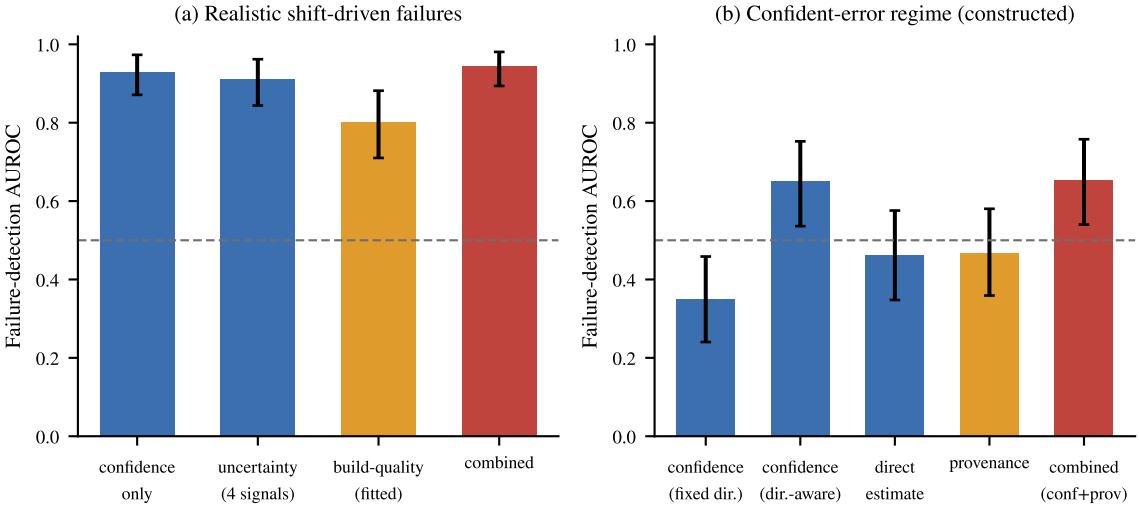
(a) Realistic shift-driven failures
Failure-detection AUROC
1.0
0.8
0.6
0.4
0.2
0.0
confidence only
uncertainty (4 signals)
build-quality (fitted)
combined
(b) Confident-error regime (constructed)
Failure-detection AUROC
1.0
0.8
0.6
0.4
0.2
0.0
confidence (fixed dir.)
confidence (dir.-aware)
direct estimate
provenance
combined (conf+prov)